4. Thymine:

- Thymine is a Pyrimidine nitrogenous base found only in DNA.
- In DNA, thymine pairs with adenine via two hydrogen bonds.

5. Uracil:

- Uracil is a Pyrimidine nitrogenous base found only in RNA.
- In RNA, uracil pairs with adenine via two hydrogen bonds.

Exercise :22

Types of tissues.

Aim : To study and identify the types of plant tissues and its diversity in shapes and sizes of plant cells.

Principle: Tissue is a group of cells performing a common function. Tissue may be simple (parenchyma, collenchymas and sclerenchyma) i.e., containing only one type of cells or complex (xylem, phloem) that is containing more than one type of cells. The tissues are also classified into meristematic and permanent tissues. Cells of different types of tissues differ in their structure, shape, size, function and wall composition.

Requirements: Model / photograph / pictures of aerenchyma, chlorenchyma, angular collenchyma, brachysclereid, xylem trachieds, xylem vessels, phloem sieve tube.

Diagnostic features:

1. Aerenchyma:

- Parenchyma cells with large intercellular spaces and filled with air found in aquatic plants.
- It provides buoyancy to the plants in water.

2. Chlorenchyma:

- Parenchyma cells with chloroplast found in green leaves and stem.
- The main function is to perform photosynthesis.

3. Angular collenchyma:

- Collenchyma cells have thickening at the angles where cells meet.
- It is found in hypodermis of *Datura* and *Nicotiana*.

4. Brachysclereid:

- Brachysclereids or stone cells are isodiametric with hard cell wall.
- It is found in bark, pith, cortex, hard endosperm and pulp of pyrus.

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